

A dimension-free partial answer to the affine projection question on ℓ_p^n balls

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Source and target question

The source is Tuomas Hytonen and Assaf Naor, *Heat flow and quantitative differentiation*, arXiv:1608.01915. In Section 1.F.1, Question 19 asks whether the affine orthogonal projection on an isotropic convex body is bounded on Lipschitz maps. In the source notation, if $(X, \|\cdot\|_X)$ is n -dimensional, B_X is in isotropic position with isotropic constant L_X , and Proj is the $L_2(B_X)$ -orthogonal projection onto affine functions,

$$\text{Proj}f(x) = \int_{B_X} f(z) dz + \frac{1}{L_X^2} \sum_{j=1}^n x_j \int_{B_X} z_j f(z) dz,$$

the question is whether, for every Banach space Y and every Lipschitz $f : B_X \rightarrow Y$,

$$\|\text{Proj}f\|_{\text{Lip}(B_X, Y)} \lesssim \|f\|_{\text{Lip}(B_X, Y)}$$

with a universal implicit constant. The source also records that the problem is equivalent to a scalar Wasserstein symmetry estimate.

a version of (12) with weaker dependence on n that is insufficient for proving Theorem 2 (a version of (12) that leads to a bound similar to (24) might be within reach through such an approach).

Our estimate (12) is also a quantitative improvement of (26) because, as we recalled in the end of Section 1.B.1, there exists a universal constant $\kappa \in (0, \infty)$ such that when Y is a UMD Banach space with $\beta = \beta(Y)$, if we set $q = \kappa\beta$ then we have $\mathfrak{m}_q(Y) \lesssim \beta^2$. Consequently, it follows formally from (12) that a variant of (26) holds true (using intrinsic averaging; see Section 1.F.1 below) with the quantity $n^{5/2}\beta(Y)^{15}$ in the right hand side replaced by the smaller quantity $n^{9/20}\sqrt[5]{\log n} \cdot \beta^2$.

1.F.1. Intrinsic averages. A convenient advantage of (12) over (26) is that the averaging in the left hand side of (12) occurs over the intrinsic balls $x + tB_X$ while the averaging in (26) is over the auxiliary Euclidean balls $x + tB^n$. This difference reflects a conceptual rather than technical geometric difficulty that arose in [40] and is circumvented here altogether due to the use of the heat semigroup rather than a more complicated (but natural) operator that was used in [40]. As we have seen in Section 1.D, the fact that we can work here with averages over intrinsic balls leads to a quick and direct deduction of our bound on $r^{X \rightarrow Y}(\varepsilon)$, while the fact that in (26) the averages are over Euclidean balls requires an additional argument that is carried out in [40, Section 3.1] in order to relate (26) to the uniform approximation by affine property with the stated asymptotic dependence on n . In [40] (as well as in [27, 33, 25]) the affine function Λ that is used to approximate f on the ball $x + tB^n$ is $(\text{Proj} \otimes \text{Id}_Y)f$, where Id_Y is the identity on Y and Proj is the orthogonal projection from $L_2(x + tB^n)$ onto its linear subspace $\mathcal{A}(X, \mathbb{R}) \cap L_2(x + tB^n)$ that consists of all the restrictions to $x + uB^n$ of affine functions from $\mathbb{R}^n$ to $\mathbb{R}$. Lemma 10 of [40] asserts that $\|(\text{Proj} \otimes \text{Id}_Y)f\|_{\text{Lip}(x+tB^n, Y)} \leq \|f\|_{\text{Lip}(x+tB^n, Y)}$, but its proof uses the rotational symmetry of the Euclidean ball $x + tB^n$ and the analogous statement is unknown with $x + tB^n$ replaced by $x + tB_X$. The need to address such issues does not arise in our approach because the Lipschitz constant of a heat evolute can be easily bounded without any need for additional symmetries of B_X . Nevertheless, it would be independently interesting to understand Question 19 below since the operator Proj is a natural object whose Lipschitz properties are equivalent to the potential availability of certain approximate distributional symmetries of high dimensional centrally symmetric convex bodies.

Question 19. Let $(X, \|\cdot\|_X)$ be an n -dimensional normed space, equipped with a Hilbertian norm $|\cdot|$ with respect to which it is isotropic with isotropic constant L_X , i.e., $|B_X| = 1$ and (10) holds true. Consider the orthogonal projection Proj from $L_2(B_X)$ onto the subspace of affine mappings $\mathcal{A}(X, \mathbb{R}) \cap L_2(B_X)$. Thus, for every $f \in L_2(B_X)$ and $x \in X$ we have

$$\text{Proj}f(x) = \int_{B_X} f(z) dz + \frac{1}{L_X^2} \sum_{j=1}^n x_j \int_{B_X} z_j f(z) dz. \quad (27)$$

The formula (27) also makes sense when $f \in L_1(B_X; Y)$ for any Banach space $(Y, \|\cdot\|_Y)$, and we shall use the notation $\text{Proj}f$ in this case as well (i.e., slightly abusing notation by identifying $\text{Proj} \otimes \text{Id}_Y$ with Proj). Is it true that for every Banach space Y and every Lipschitz function $f : B_X \rightarrow Y$ we have $\|\text{Proj}f\|_{\text{Lip}(B_X, Y)} \lesssim \|f\|_{\text{Lip}(B_X, Y)}$?

In Section 5.A we show that it suffices to treat the above question when $Y = \mathbb{R}$, i.e., the following operator norm identity holds true for every Banach space $(Y, \|\cdot\|_Y)$.

$$\|\text{Proj}\|_{\text{Lip}(B_X, Y) \rightarrow \text{Lip}(B_X, Y)} = \|\text{Proj}\|_{\text{Lip}(B_X, \mathbb{R}) \rightarrow \text{Lip}(B_X, \mathbb{R})}. \quad (28)$$

Moreover, the quantity $\|\text{Proj}\|_{\text{Lip}(B_X, \mathbb{R}) \rightarrow \text{Lip}(B_X, \mathbb{R})}$ has the following geometric interpretation.

For every $x \in X \setminus \{0\}$ let μ_x^+, μ_x^- be the probability measures supported on B_X whose densities are given for every $y \in X$ by

$$d\mu_x^+(y) \stackrel{\text{def}}{=} \frac{\max\{x \cdot y, 0\} \mathbf{1}_{B_X}(y)}{\frac{1}{2} \int_{B_X} |x \cdot z| dz} dy \quad \text{and} \quad d\mu_x^-(y) \stackrel{\text{def}}{=} \frac{\max\{-x \cdot y, 0\} \mathbf{1}_{B_X}(y)}{\frac{1}{2} \int_{B_X} |x \cdot z| dz} dy. \quad (29)$$

Result

This packet proves a dimension-free affirmative answer for the classical finite-dimensional ℓ_p balls.

Theorem 1. Fix $p \in [1, \infty]$. For every $n \in \mathbb{N}$, let $X = \ell_p^n$, and put its unit ball in the volume-normalized isotropic position used in Question 19 of Hytonen–Naor. Let Y be any Banach space.

Then the affine projection Proj satisfies

$$\|\text{Proj}f\|_{\text{Lip}(B_{\ell_p^n}, Y)} \leq C_p \|f\|_{\text{Lip}(B_{\ell_p^n}, Y)}$$

for every Lipschitz map $f : B_{\ell_p^n} \rightarrow Y$, where C_p is independent of n . For $p = \infty$ one may take $C_\infty = 3/2$. For each finite p , one may take

$$C_p = \sup_{n \geq 1} \frac{1}{2\sigma_{p,n}^2} \left(\frac{\Gamma(3 + 1/p)\Gamma(1 + n/p)}{\Gamma(1 + 1/p)\Gamma(3 + n/p)} \right)^{1/p},$$

where

$$\sigma_{p,n}^2 = \frac{\Gamma(3/p)\Gamma(1 + n/p)}{\Gamma(1/p)\Gamma(1 + (n + 2)/p)}.$$

The displayed supremum is finite for every fixed $p < \infty$.

Proof intuition

For a centrally symmetric body the affine projection is controlled, in the scalar case, by covariances of the form $\int z_j f(z) dz$. On an ℓ_p ball, each such covariance has a one-dimensional integration-by-parts representation along the j th coordinate:

$$\int z_j f(z) dz = \int W_j(z) \partial_j f(z) dz,$$

where W_j is the gap between the square of the coordinate slice radius and z_j^2 . If scalar f is 1-Lipschitz for the ℓ_p metric, then $|\nabla f|_{\ell_{p'}} \leq 1$ a.e. Thus a linear combination of the covariance terms is bounded by an L_p moment of the weights W_j . That moment is of the same order as the coordinate variance of the uniform measure on the ℓ_p ball, uniformly in the dimension. Dividing by the covariance normalization in the projection formula gives the dimension-free scalar bound; the Hytonen–Naor scalar-to-Banach identity then gives the stated result for every target Y .

Proof

By equation (28) in the source paper, the Lipschitz operator norm of Proj with arbitrary Banach target Y is equal to its scalar-valued Lipschitz operator norm. It is therefore enough to prove the theorem for scalar-valued Lipschitz functions. We work with the standard unit ball

$$K_p^n = \{z \in \mathbb{R}^n : \|z\|_p \leq 1\}$$

and normalized Lebesgue probability measure on it. A common dilation of the body changes the covariance and the Minkowski functional in opposite ways and leaves the Lipschitz operator norm of Proj unchanged. The standard Euclidean structure is isotropic for K_p^n by sign and permutation symmetry, after volume normalization.

Let Z be uniformly distributed on K_p^n . For $p < \infty$, write

$$\sigma_{p,n}^2 = \mathbb{E}Z_1^2.$$

For smooth scalar $f : K_p^n \rightarrow \mathbb{R}$, the affine projection has linear part

$$h \mapsto \frac{1}{\sigma_{p,n}^2} \mathbb{E}[(h \cdot Z)f(Z)].$$

Smoothness is harmless: the general scalar Lipschitz case follows by the usual mollification argument on slightly smaller balls and passage to the limit, or equivalently by applying the following integration-by-parts identities to a.e. classical derivatives.

Fix j and freeze all coordinates except z_j . If

$$A_j(z) = \left(1 - \sum_{i \neq j} |z_i|^p\right)^{1/p},$$

then along that slice $z_j \in [-A_j, A_j]$. Integration by parts gives

$$\int_{-A_j}^{A_j} t f(\dots, t, \dots) dt = \frac{1}{2} \int_{-A_j}^{A_j} (A_j^2 - t^2) \partial_j f(\dots, t, \dots) dt.$$

Consequently

$$\mathbb{E}[Z_j f(Z)] = \mathbb{E}[W_j(Z) \partial_j f(Z)], \quad W_j(Z) = \frac{A_j(Z)^2 - Z_j^2}{2}.$$

Let $h \in \ell_p^n$. Since $|\nabla f(Z)|_{\ell_p^n} \leq \|f\|_{\text{Lip}}$ almost everywhere,

$$\begin{aligned} |\mathbb{E}[(h \cdot Z) f(Z)]| &= \left| \mathbb{E} \sum_{j=1}^n h_j W_j(Z) \partial_j f(Z) \right| \\ &\leq \|f\|_{\text{Lip}} \mathbb{E} \left(\sum_{j=1}^n |h_j|^p W_j(Z)^p \right)^{1/p} \\ &\leq \|f\|_{\text{Lip}} \|h\|_p (\mathbb{E} W_1^p)^{1/p}. \end{aligned}$$

Here the last step uses Jensen's inequality and symmetry of the W_j . Thus

$$\|\text{Proj} f\|_{\text{Lip}(K_p^n, Y)} \leq \frac{(\mathbb{E} W_1^p)^{1/p}}{\sigma_{p,n}^2} \|f\|_{\text{Lip}(K_p^n, Y)}.$$

For $p = 1$ this already yields the sharper bound 1, because $\mathbb{E} W_1 = \mathbb{E} Z_1^2$.

For $1 < p < \infty$, use $0 \leq W_1 \leq A_1^2/2$. A beta-integral calculation gives

$$\mathbb{E} A_1^{2p} = \frac{\Gamma(3 + 1/p) \Gamma(1 + n/p)}{\Gamma(1 + 1/p) \Gamma(3 + n/p)}$$

and

$$\sigma_{p,n}^2 = \mathbb{E} Z_1^2 = \frac{\Gamma(3/p) \Gamma(1 + n/p)}{\Gamma(1/p) \Gamma(1 + (n+2)/p)}.$$

Therefore

$$\frac{(\mathbb{E} W_1^p)^{1/p}}{\sigma_{p,n}^2} \leq \frac{1}{2\sigma_{p,n}^2} \left(\frac{\Gamma(3 + 1/p) \Gamma(1 + n/p)}{\Gamma(1 + 1/p) \Gamma(3 + n/p)} \right)^{1/p}.$$

The standard gamma-ratio estimate $\Gamma(t+a)/\Gamma(t+b) \asymp_{a,b} t^{a-b}$ shows that this expression is bounded in n for each fixed $p < \infty$.

It remains to record the endpoint $p = \infty$. Now $K_\infty^n = [-1, 1]^n$ and $\sigma_{\infty,n}^2 = \mathbb{E}Z_1^2 = 1/3$. One-dimensional integration by parts gives

$$\mathbb{E}[Z_j f(Z)] = \mathbb{E}\left[\frac{1 - Z_j^2}{2} \partial_j f(Z)\right].$$

For $h \in \ell_\infty^n$,

$$|\mathbb{E}[(h \cdot Z) f(Z)]| \leq \|f\|_{\text{Lip}} \mathbb{E} \left\| \left(h_j \frac{1 - Z_j^2}{2} \right)_{j=1}^n \right\|_\infty \leq \frac{1}{2} \|h\|_\infty \|f\|_{\text{Lip}}.$$

After division by $\sigma_{\infty,n}^2 = 1/3$, this gives

$$\|\text{Proj} f\|_{\text{Lip}(K_\infty^n, Y)} \leq \frac{3}{2} \|f\|_{\text{Lip}(K_\infty^n, Y)}.$$

This proves the theorem.

Verification and limitations

The proof is exact for the whole ℓ_p^n scale and all Banach targets Y . It does not answer Question 19 for arbitrary isotropic convex bodies. The constants for finite p are not optimized; for instance, the Euclidean ball has a better bound by rotational symmetry.

The script `code/check_gamma_constants.py` numerically samples the explicit finite- p Gamma-ratio bound for representative p and n . This is only a sanity check; the dimension-free assertion follows from the displayed Gamma-ratio asymptotics.

Novelty check: the cheap run indexes had no hit for arXiv:1608.01915, the paper title, or distinctive ‘Q:proj lip’ phrases. A bounded web search for exact affine-projection and Wasserstein-symmetry phrases found the source arXiv page but no later direct answer. This is therefore recorded as a likely-new partial result, subject to human review.

References

- T. Hytonen and A. Naor, *Heat flow and quantitative differentiation*, arXiv:1608.01915, 2016.